

ASYMPTOTIC RS MCA

ABSTRACT. We give a compact proof of the asymptotic Reed–Solomon mutual-correlated-agreement frontier from the paid ledger developed in the v13 raw and Grande Finale manuscripts. The proof reduces the remaining primitive prefix problem to a fixed-density Boolean slice mapped by Vandermonde moments. After the Fourier/Sidon cell is paid, any positive logarithmic moment excess gives a large high-energy Boolean fiber. Balog–Szemerédi–Gowers and quasicycle difference growth forbid such a fiber, so primitive Q holds with only $\exp(o(n))$ loss. Combined with the identity-prefix lower construction, every smooth row sequence covered by the closed ledger has threshold $1 - \rho - g^*(\rho, \log_2 |\mathbb{B}|) + o(1)$. The closed-ledger package is stated explicitly, with the individual cell payments cited to the existing ledgers.

1. STATEMENT

Let $D \subseteq \mathbb{F}$ be an evaluation set of size n , and let

$$C = \text{RS}_{\mathbb{F}}(D, k) = \{(f(t))_{t \in D} : \deg f < k\}.$$

For $a \in \{0, \dots, n\}$, a finite slope $\gamma \in \mathbb{F}$ is support-wise MCA-bad for a pair $r_1, r_2 \in \mathbb{F}^D$ at agreement a if some set $S \subseteq D$, $|S| \geq a$, and some $h \in \mathbb{F}[X]$, $\deg h < k$, satisfy $r_1 + \gamma r_2 = h$ on S , while r_1, r_2 are not simultaneously explained on the same S by two degree- $< k$ polynomials. Put

$$B_C^{\text{MCA}}(a) = \max_{r_1, r_2} \#\{\gamma \in \mathbb{F} : \gamma \text{ is MCA-bad at agreement } a\}.$$

Then $\varepsilon_{\text{mca}}(C, 1 - a/n) = B_C^{\text{MCA}}(a)/q_{\text{line}}$, where q_{line} is the line-slope field used by the MCA sampler. This integer staircase convention is the one used in the v13 raw ledger [Cho26a] and in the Grande Finale reduction [Cho26b].

We consider smooth multiplicative or circle-type RS row sequences $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ with $|D_n| = n$, $k_n/n \rightarrow \rho \in (0, 1)$, and a base/generated alphabet $\mathbb{B}_n$ of bit-size $\beta_n = \log_2 |\mathbb{B}_n| \rightarrow \beta$. For an agreement $a_n = k_n + 1 + w_n$, define the prefix average

$$\overline{N}_{n, a_n} = \binom{n}{a_n} |\mathbb{B}_n|^{-w_n}.$$

The entropy–subfield envelope is

$$g^*(\rho, \beta) = \sup\{g \in [0, 1 - \rho] : H_2(\rho + g) \geq \beta g\},$$

where $H_2(x) = -x \log_2 x - (1 - x) \log_2 (1 - x)$. The target threshold is $\delta_C^*(\varepsilon) = \sup\{\delta : \varepsilon_{\text{mca}}(C, \delta) \leq \varepsilon\}$.

Definition 1.1 (Cells and first-match payment). *A cell is a constructible subfamily of the MCA witness incidence space, or a bounded-complexity family of such subfamilies, stable under extension of the ground field. Cells are ordered. A slope is assigned to the first cell containing one of its witnesses. The cell is paid at agreement a_n if the number of slopes assigned to it is at most its printed budget, and a family is paid if the sum of its first-match budgets is $\exp(o(n))\overline{N}_{n, a_n}$.*

Lemma 1.2 (First-match disjointization). *If ordered cells $C_1, \dots, C_J$ cover a set of bad slopes and the j -th first-match cell has budget U_j , then the covered slopes contribute at most $\sum_j U_j$ to $B_C^{\text{MCA}}(a)$.*

Proof. For a fixed received line, assign each bad slope to the least-indexed cell containing a witness. These assigned slope classes are disjoint, and the j -th class is contained in the projection of C_j after earlier cells have been removed. Summing the budgets and then maximizing over received lines proves the claim. $\square$

Definition 1.3 (Closed paid ledger). *A smooth RS row sequence is closed-ledger at agreements a_n if the following first-match cells cover all MCA-bad witnesses except the primitive leaves of Definition 2.1, and if their total contribution is $\exp(o(n))\bar{N}_{n,a_n}$.*

- (C1) Quotient-pullback cells. *The support or locator descends along a nontrivial finite map $D \rightarrow D'$. These cells include quotient-remainder locators, divisor-block support ledgers, quotient-image lcm ledgers, and quotient pullback recursion.*
- (C2) Chebyshev and dihedral cells. *On multiplicative or circle rows, the support is invariant under the dihedral action, or the locator factors through a Chebyshev or circle quotient such as $z + z^{-1}$.*
- (C3) Planted-block cells. *A positive-density block is forced into the support or common zero set. The payment is the entropy cost of planting that block, together with the bounded quotient profile.*
- (C4) Tangent and deep-center cells. *The received line lies in a tangent, secant, or high-contact stratum of the RS code variety, including the high-agreement tangent cell.*
- (C5) Extension and descent cells. *The witness either descends to a smaller field or is controlled by a bounded extension-line or extension-pole parameter.*
- (C6) Differential-locator and regular-rank cells. *The locator map has a rank drop, a ramification defect, or an overdetermined Hankel minor. These cells include regular closed-range Hankel packing, canonical rank-drop ledgers, and curve rank-drop ledgers.*
- (C7) Saturation and effective-image-collapse cells. *The apparent support count is larger than the actual line-ray or codeword-ray image. The payment uses exact saturation identities and exact residual image ledgers.*
- (C8) Balanced-core and split-pencil cells. *The residual witness lies in a balanced-core split-pencil chart. One-parameter primitive locator pencils are paid by the moving-root theorem; deficiency-one and identically split top charts are tangent-borne or eliminated.*
- (C9) Fourier/Sidon cells. *In a primitive prefix leaf, a heavy fiber with exponentially small additive closure is cut as a Fourier/Sidon cell: major arcs are routed to (C1)–(C8), and minor arcs are paid by Fourier flatness.*

The remaining first-match complement is a disjoint union of primitive prefix leaves. Every such leaf has the Boolean fixed-density Vandermonde form of Definition 2.1.

Theorem 1.4 (Closed ledger package). *For the smooth multiplicative and circle-type RS row sequences treated in the v13 raw ledger and in the Grande Finale notes, the cells (C1)–(C9) form a closed paid ledger in the asymptotic frontier window.*

Proof. We record the existing ledger package rather than reprove each algebro-geometric count.

For (C1), the quotient-remainder lower and support ledgers, divisor-union support ledger, exact finite-parameter quotient-image ledger, affine/projective lcm ledgers, quotient pullback recursion, and bounded quotient census are proved in [Cho26a, quotient-remainder, quotient support, quotient-image, and bounded-census ledgers]. The corresponding first-match convention and quotient status cell are isolated in [Cho26b].

For (C2), exact Chebyshev fibers on circle twin-coset domains, circle quotient transport, and circle deployed ledgers are proved in [Cho26a, Chebyshev and circle sections]. Dihedral and Chebyshev cells are quotient cells with extra monodromy and are therefore paid by the same downstairs-support and image ledgers.

For (C3), planted quotient profiles, planted quotient-core lower counts, planted list-budget windows, and bounded active quotient order are established in [Cho26a, planted quotient profile and bounded quotient census]. These supply the entropy payment for positive-density forced blocks.

For (C4), the exact high-agreement tangent cell and tangent-borne split-kernel sections are proved in [Cho26a, high-agreement tangent exact cell and split-kernel tangent-borne theorem] and refined in [Cho26b, exact high-agreement tangent cell].

For (C5), the extension-pole deep-list and quotient-remainder floors, extension-line dimension-degree ledger, subfield confinement, and genuinely extension-valued line accounting are proved in

[Cho26a, extension-pole and extension-line ledgers]; the full-extension cell target and extension-valued rank-one floor are recorded in [Cho26b].

For (C6), regular Hankel minors, canonical exact-agreement rank-drop ledgers, closed-ball rank-drop lcm ledgers, curve rank-drop ledgers, arbitrary-residual image ledgers, and the scanner residual ledger are proved in [Cho26a, Hankel, rank-drop, curve, residual-image, and scanner ledgers]. These are exactly the differential-locator and regular-rank payments.

For (C7), the exact saturation identity and line-ray saturation identity are proved in [Cho26b, saturation and line-ray saturation identities]; exact arbitrary-residual image and closed-ball residual image ledgers appear in [Cho26a, residual image ledgers]. Thus raw support overcounts are replaced by actual image counts.

For (C8), the interpolation-lattice split-pencil reduction, near-rational payment, deficiency-one split-test eliminant, split annihilator dictionary, tangent-borne split charts, boundary-profile identification with $\mathbb{Q}$, base-field split-pencil floor, one-parameter moving-root theorem, and primitive one-pencil corollary are proved in [Cho26b, split-pencil and BC sections], with the balanced-core and split-pencil reductions also developed in [Cho26a, rank-one and split-pencil reductions].

For (C9), Fourier-flat leaves satisfy asymptotic $\mathbb{Q}$ and large-characteristic Fourier examples are proved in [Cho26b, Fourier-flat $\mathbb{Q}$ section]. The Sidon/Fourier operational cut, the split between Sidon-heavy and high-energy fibers, and the fact that major arcs route to algebraic cells while minor arcs are Fourier-flat are the moduli-ledger results of [Cho26d, Cho26c]. They are restated and used below in Definition 3.1, Proposition 3.2, and Theorem 4.4.

Finally, first-match disjointization is Lemma 1.2 here and is also proved in the Grande Finale and moduli-ledger notes. The listed cells exhaust the nonprimitive branches of the v13 first-match atlas; after removing them, the remaining leaf is the primitive Boolean prefix slice. Summing the printed budgets gives $\exp(o(n))\bar{N}_{n,a_n}$, which is the closed paid ledger. $\square$

Thus the row-specific geometry is not an additional conjectural input to this paper: it is the imported closed ledger package. The remaining work below is the uniform primitive count and the compiler from that count to the MCA staircase.

Theorem 1.5 (Asymptotic RS–MCA frontier). *Let $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ be a smooth RS row sequence as above. Assume the closed paid ledger holds uniformly in every $o(n)$ -window around the crossing $a_n = (\rho + g^*)n + o(n)$. Assume also the standard identity-prefix pole construction supplies the lower side below the crossing. Then, for every target sequence with $\log_2(1/\varepsilon_n) = O(n)$ and fixed challenge normalization,*

$$\delta_{C_n}^*(\varepsilon_n) = 1 - \rho - g^*(\rho, \beta) + o(1),$$

with the evident replacement of β by $\lim \log_2 |\mathbb{B}_n|$ when the base alphabet varies.

The rest of the paper proves Theorem 1.5. The proof is self-contained after the imported closed-ledger package and the standard external additive-combinatorics facts stated in Section 4.

2. PRIMITIVE LEAVES

Definition 2.1 (Primitive prefix leaf). *A primitive leaf consists of a finite set T , $|T| = N$, an integer m with m/N bounded away from 0 and 1, a field K , nonzero weights $\rho(t) \in K^\times$, and columns*

$$v_t = \rho(t)(1, t, \dots, t^{R-1}) \in K^R, \quad t \in T.$$

The active support family is a first-match residual slice

$$\Omega^\circ \subseteq \{x \in \{0, 1\}^T : \sum_{t \in T} x_t = m\},$$

and the prefix map is $\Phi(x) = \sum_{t \in T} x_t v_t$. Write $\mathcal{S} = \text{im } \Phi$, $L = |\mathcal{S}|$, $M = |\Omega^\circ|$, $\bar{N} = M/L$, and $F_s = \Omega^\circ \cap \Phi^{-1}(s)$. The leaf is at the frontier if $\log \bar{N} = o(N)$.

For $q = q_N \rightarrow \infty$, define

$$\Gamma_q^{\text{ord}} = L^{-1} \sum_{s \in S} \left(\frac{|F_s|}{\bar{N}} \right)^q.$$

Lemma 2.2 (Moment-max equivalence). *If $q \rightarrow \infty$ and $\log L = o(Nq)$, then*

$$L^{-1} \left(\max_s \frac{|F_s|}{\bar{N}} \right)^q \leq \Gamma_q^{\text{ord}} \leq \left(\max_s \frac{|F_s|}{\bar{N}} \right)^q.$$

Hence $\Gamma_q^{\text{ord}} \leq \exp(o(Nq))$ is equivalent to $\max_s |F_s| \leq \exp(o(N))\bar{N}$.

Proof. The upper bound replaces every summand by the maximum; the lower bound keeps a maximum summand. Taking q -th roots and using $\log L = o(Nq)$ gives the final assertion. $\square$

3. SIDON CUT

For a finite F in an abelian group, let

$$E(F) = \#\{(a, b, c, d) \in F^4 : a - b = c - d\}, \quad \Delta(F) = E(F)/|F|^3.$$

Thus $\Delta(F)$ is the probability that $a - b + c \in F$ for independent uniform $a, b, c \in F$.

For $\sigma > 0$, define the Sidon-heavy moment of a primitive leaf by

$$\Gamma_{q,\sigma}^{\text{sid}} = L^{-1} \sum_{\Delta(F_s) \leq e^{-\sigma N}} \left(\frac{|F_s|}{\bar{N}} \right)^q.$$

Definition 3.1 (Paid Fourier/Sidon cell). *The Fourier/Sidon cell is paid if, for every fixed $\sigma > 0$ and every logarithmic q , one has $\Gamma_{q,\sigma}^{\text{sid}} \leq \exp(o(Nq))$. Equivalently, no positive-rate heavy fiber with exponentially small additive closure remains in the primitive residual.*

In practice this cell is paid by Fourier flatness after major arcs have been routed to algebraic cells. If $\mu = \Phi_* \text{Unif}(\Omega^\circ)$ and $\sum_{\chi \neq 1} |\hat{\mu}(\chi)| \leq \exp(o(N))$, Fourier inversion gives $L\mu(s) \leq \exp(o(N))$ for all s , hence primitive $\mathbf{Q}$. In RS moment-curve leaves, large Fourier coefficients are algebraic major arcs: quotient, descent, dihedral, ramification, or effective-image-collapse phenomena. These are exactly the earlier paid cells in Definition 1.3.

Proposition 3.2 (Energy extraction after the Sidon cut). *Assume the Fourier/Sidon cell is paid. If $\Gamma_q^{\text{ord}} \geq e^{\eta Nq}$ for some fixed $\eta > 0$, then some level $s = s_N$ satisfies $|F_s| \geq e^{\eta N - o(N)}\bar{N}$ and $E(F_s) \geq |F_s|^3 e^{-o(N)}$.*

Proof. Fix $\sigma > 0$. Split Γ_q^{ord} into terms with $\Delta(F_s) \leq e^{-\sigma N}$ and the complementary terms. The first part is $\Gamma_{q,\sigma}^{\text{sid}} = \exp(o(Nq))$, so a complementary term has $(|F_s|/\bar{N})^q \geq e^{\eta Nq - o(Nq)}$. Hence $|F_s| \geq e^{\eta N - o(N)}\bar{N}$ and $\Delta(F_s) > e^{-\sigma N}$. Letting $\sigma \downarrow 0$ slowly along the sequence gives $E(F_s) \geq |F_s|^3 e^{-o(N)}$. $\square$

4. BOOLEAN ADDITIVE COMBINATORICS

We use the energy form of Balog–Szemerédi–Gowers [BS94, Gow98]. Its precise polynomial exponents are irrelevant here.

Theorem 4.1 (Balog–Szemerédi–Gowers). *If A is a finite subset of an abelian group and $E(A) \geq |A|^3/K$, then some $A' \subseteq A$ satisfies $|A'| \geq K^{-C}|A|$ and $|A' - A'| \leq K^C|A'|$, where C is absolute.*

We also use the quasicube sumset theorem of Matolcsi–Ruzsa–Shakan–Zhelezov [MRSZ20] in the sharpened form of Green–Matolcsi–Ruzsa–Shakan–Zhelezov [GMRSZ20]: if $U \subseteq \{0, 1\}^d$, then $|P + Q + U| \geq |P|^{1/2}|Q|^{1/2}|U|$ for finite nonempty $P, Q \subseteq \mathbb{Z}^d$.

Theorem 4.2 (Quasicube difference growth). *Every finite $A \subseteq \{0, 1\}^N$ satisfies $|A - A| \geq |A|^{3/2}$.*

Proof. Apply the quasicube theorem with $U = A$, $P = -A$, and $Q = \{0\}$. $\square$

Proposition 4.3 (No large high-energy Boolean fiber). *There is no sequence $F_N \subseteq \{0, 1\}^N$ with $|F_N| \geq e^{cN-o(N)}$ and $E(F_N) \geq |F_N|^3 e^{-o(N)}$ for fixed $c > 0$.*

Proof. By Theorem 4.1 with $K = e^{o(N)}$, a subset $F'_N \subseteq F_N$ has $|F'_N| \geq e^{cN-o(N)}$ and $|F'_N - F'_N| \leq e^{o(N)}|F'_N|$. But Theorem 4.2 gives $|F'_N - F'_N| \geq |F'_N|^{3/2}$, so $|F'_N|^{1/2} \leq e^{o(N)}$, a contradiction. $\square$

Theorem 4.4 (Primitive Q after the Sidon cut). *In every frontier primitive leaf with paid Fourier/Sidon cell, $\max_s |F_s| \leq \exp(o(N))\bar{N}$.*

Proof. If not, Lemma 2.2 gives $\Gamma_q^{\text{ord}} \geq e^{\eta Nq}$ for some fixed $\eta > 0$ along a subsequence and some logarithmic q . Proposition 3.2 then gives a fiber $F_s \subseteq \{0, 1\}^T$ with $|F_s| \geq e^{\eta N-o(N)}\bar{N} = e^{cN-o(N)}$ and $E(F_s) \geq |F_s|^3 e^{-o(N)}$, because $\log \bar{N} = o(N)$. This contradicts Proposition 4.3. $\square$

5. MCA COMPILER

Lemma 5.1 (First-match add-back). *For a closed-ledger row, the full first-match prefix count after paid cells satisfies $\max_s N(s) \leq \exp(o(n))\bar{N}_{n,a_n}$.*

Proof. Paid cells contribute $\exp(o(n))\bar{N}_{n,a_n}$ by closure. Primitive leaves satisfy Theorem 4.4; their normalized averages sum to at most $\exp(o(n))\bar{N}_{n,a_n}$ by the first-match profile decomposition. Summing over the subexponential profile family preserves the same scale. $\square$

Lemma 5.2 (Q pays shift pairs). *If $\sum_s N(s) = M$, $\bar{N} = M/L$, and $\max_s N(s) \leq \kappa\bar{N}$, then $M^{-1} \sum_s N(s)^2 \leq \kappa\bar{N}$.*

Proof. $\sum_s N(s)^2 \leq (\max_s N(s)) \sum_s N(s) \leq \kappa\bar{N}M$. $\square$

Theorem 5.3 (Closed-ledger upper bound). *If a smooth RS row sequence is closed-ledger at a_n , then*

$$B_{C_n}^{\text{MCA}}(a_n) \leq \exp(o(n)) \binom{n}{a_n} |\mathbb{B}_n|^{-(a_n-k_n-1)}.$$

Proof. The nonprimitive cells are paid by definition. The primitive prefix leaves obey Q by Theorem 4.4 and add back by Lemma 5.1. The shift-pair second-moment ledger is paid by Lemma 5.2. First-match disjointization sums these disjoint slope classes, giving the displayed numerator bound. $\square$

6. LOWER SIDE AND FRONTIER

We recall the identity-prefix pole construction in the form needed here. Let $m = k+1+w$, and for $S \subseteq D$, $|S| = m$, write $\ell_S(X) = \prod_{t \in S} (X - t)$. Pigeonholing the first w locator coefficients over $\mathbb{B}^w$ gives a prefix vector z with at least $\binom{n}{m} |\mathbb{B}|^{-w}$ supports. Choose a pole $\alpha \in \mathbb{F} \setminus D$, and set $U_z(X) = X^m + z_1 X^{m-1} + \dots + z_w X^{m-w}$. The line

$$f_\alpha(x) = \frac{U_z(x)}{x - \alpha}, \quad g_\alpha(x) = \frac{-1}{x - \alpha}, \quad x \in D,$$

has the property that $f_\alpha + \zeta g_\alpha$ is code-explained on S exactly when $\ell_S(\alpha) = U_z(\alpha) - \zeta$. The word g_α is not explained by a degree- $< k$ polynomial on $k+1$ positions, so the corresponding slopes are MCA-bad. Except for the usual collision loss in evaluating $\ell_S(\alpha)$, which is subexponential in the standard pole-reservoir regime, this gives

$$B_C^{\text{MCA}}(m) \geq \exp(-o(n)) \binom{n}{m} |\mathbb{B}|^{-w}.$$

This is the lower construction used in [Cho26a, Cho26b].

Proof of Theorem 1.5. Take $a_n = (\rho + g)n + o(n)$. Stirling gives

$$\log_2 \bar{N}_{n,a_n} = n(H_2(\rho + g) - \beta g) + o(n).$$

If $g > g^*(\rho, \beta)$ by a fixed amount, this exponent is negative by $\Omega(n)$, so Theorem 5.3 makes $B_{C_n}^{\text{MCA}}(a_n)$ smaller than any target with the prescribed normalization after moving $o(n)$ agreements to absorb the $\exp(o(n))$ ledger overhead. Thus the safe radius reaches $1 - \rho - g^*(\rho, \beta) - o(1)$. If $g < g^*(\rho, \beta)$ by a fixed amount, the identity-prefix construction gives exponentially many bad slopes, so the target fails at radius $1 - \rho - g + o(1)$. Letting g approach g^* from both sides proves the claimed equality. $\square$

Remark 6.1 (Finite adjacent rows). This is an asymptotic theorem. It intentionally absorbs $\exp(o(n))$ losses. The finite deployed adjacent pairs in [Cho26a, Cho26b] require the same first-match ledger with printed constants for every paid cell, Fourier/Sidon tail, and add-back step; those constants are not supplied by the asymptotic argument alone.

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